

Fashion Category Investment Intelligence Platform

Comprehensive Data Model

Market & Signals 5 entities	Portfolio & Deals 5 entities	Brand & Product 5 entities	Consumer & Sentiment 3 entities	Intelligence Engine 5 entities
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23 entities · 5 domain layers · 200+ fields
Version 1.0 · May 2026

Architecture Overview

The Fashion Category Investment Intelligence Platform is structured across five domain layers that flow from raw market signals through to scored investment recommendations. Each layer produces entities that feed the next, culminating in versioned InvestmentScores and audit-ready ReportSnapshots.

Layer	Entiti es	Purpose
Market & Signals	5	Core taxonomy, real-time feed ingestion, trend indicators, and macro-economic context. Provides the demand-sid...
Portfolio & Deals	5	Investment ledger covering fund structures, deal transactions, point-in-time valuations, modelled exit paths, ...
Brand & Product	5	The investable asset layer: brand financial KPIs, product line sell-through, price architecture, channel mix, ...
Consumer & Sentiment	3	Demand-side intelligence: psychographic segments, NLP-derived sentiment aggregation, and platform-level social...
Intelligence Engine	5	The scoring, forecasting, alerting, and reporting layer. Aggregates all upstream signals into versioned Invest...

Field Tag Legend

PK	Primary key	FK	Foreign key reference	IDX	Indexed for query performance	CALC	Derived / computed field	ARR	Array / multi-value field
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Market & Signals

Core taxonomy, real-time feed ingestion, trend indicators, and macro-economic context. Provides the demand-side intelligence foundation for all scoring and forecasting.

FashionCategory

Top-level taxonomy node supporting nested sub-categories (via self-referential FK). Stores TAM and 5-year CAGR as computed fields.

field	type	tag
category_id	uuid	PK
name	varchar(100)	
slug	varchar(100)	IDX
parent_category_id	uuid	FK
level	int	
tam_usd_bn	decimal(12, 2)	CALC
cagr_pct_5yr	decimal(5, 2)	CALC
seasonality_index	jsonb	
created_at	timestampz	
metadata	jsonb	
relations: MarketSegment → TrendIndicator → Brand → InvestmentScore		

MarketSegment

Geographic, demographic, or occasion-based sub-slice of a category with revenue and growth tracking.

field	type	tag
segment_id	uuid	PK
category_id	uuid	FK
geography_iso	char(3)	IDX
channel_type	enum	
price_band	enum	
revenue_usd_mn	decimal(14, 2)	CALC
yoy_growth_pct	decimal(6, 3)	CALC
penetration_rate	decimal(5, 4)	
as_of_date	date	IDX
relations: MacroEconomicData → ConsumerSegment → SentimentSignal		

MarketSignal		
Real-time data points from external feeds — sell-through rates, search volume, social velocity — with confidence scoring.		
field	type	tag
signal_id	uuid	PK
category_id	uuid	FK
signal_type	enum	IDX
source	varchar(80)	
value	decimal(14, 4)	
unit	varchar(30)	
confidence_score	decimal(4,3)	CALC
captured_at	timestamp z	IDX
raw_payload	jsonb	
is_validated	bool	
relations: TrendIndicator → AlertRule → ForecastModel		

TrendIndicator		
Derived composite metric per category across dimensions: momentum, maturity, disruption risk, and sustainability relevance.		
field	type	tag
indicator_id	uuid	PK
category_id	uuid	FK
dimension	enum	
score	decimal(5,2)	CALC
direction	enum	
signal_ids	uuid[]	ARR
model_version	varchar(20)	
computed_at	timestamp z	IDX
ttl_hours	int	
relations: InvestmentScore → Benchmark → ForecastModel		

MacroEconomicData		
CPI, disposable income, FX rates, and consumer confidence by region — ingested from public statistical agencies.		
field	type	tag
macro_id	uuid	PK
indicator_code	varchar(30)	IDX
geography_iso	char(3)	IDX
value	decimal(14, 4)	
period	date	IDX
frequency	enum	
source_org	varchar(60)	
revision_flag	bool	
relations: ForecastModel → InvestmentScore		

Portfolio & Deals

Investment ledger covering fund structures, deal transactions, point-in-time valuations, modelled exit paths, and investor mandate profiles.

Portfolio

Investment vehicle or fund grouping deals and theses. Tracks committed vs deployed capital and target IRR against category mandates.

field	type	tag
portfolio_id	uuid	PK
name	varchar(120)	
fund_type	enum	
vintage_year	int	
committed_capital_usd	decimal(16,2)	
deployed_capital_usd	decimal(16,2)	CALC
target_irr_pct	decimal(5,2)	
category_focus_ids	uuid[]	ARR
investor_id	uuid	FK
created_at	timestampz	
relations: Deal → InvestorProfile → Benchmark		

Deal

Individual investment transaction in a fashion brand. Central transactional entity linking Brand to Portfolio with full financial tracking.

field	type	tag
deal_id	uuid	PK
portfolio_id	uuid	FK
brand_id	uuid	FK
deal_type	enum	
entry_date	date	IDX
entry_valuation_usd	decimal(16, 2)	
equity_stake_pct	decimal(5, 2)	
invested_amount_usd	decimal(16, 2)	
current_moic	decimal(6, 3)	CALC
status	enum	IDX
exit_date	date	
exit_valuation_usd	decimal(16, 2)	
investment_thesis_id	uuid	FK
relations: Valuation → ExitScenario → InvestmentScore → Brand		

Valuation

Point-in-time fair value estimate with methodology (DCF, comps, precedent transactions). Includes analyst and approver tracking.

field	type	tag
valuation_id	uuid	PK
deal_id	uuid	FK
valuation_date	date	IDX
enterprise_value_usd	decimal(16, 2)	
ev_revenue_multiple	decimal(6, 2)	CALC
ev_ebitda_multiple	decimal(6, 2)	CALC
method	enum	
comps_used	uuid[]	ARR
discount_rate_pct	decimal(5, 2)	
analyst_id	uuid	FK
approved_by	uuid	FK
relations: Benchmark → ForecastModel → ReportSnapshot		

ExitScenario		
Modelled exit paths (strategic sale, IPO, secondary) with probability weights, projected IRR and MOIC.		
field	type	tag
scenario_id	uuid	PK
deal_id	uuid	FK
scenario_type	enum	
exit_horizon_months	int	
projected_ev_usd	decimal(16, 2)	
irr_pct	decimal(6,2)	CALC
moic	decimal(6,3)	CALC
probability_weight	decimal(4,3)	
assumptions_json	jsonb	
created_at	timestampz	
relations: ForecastModel → ReportSnapshot		

InvestorProfile		
LP, GP, or co-investor entity with AUM, category mandates, geography focus, and ESG priority flags.		
field	type	tag
investor_id	uuid	PK
name	varchar(140)	
type	enum	
aum_usd_bn	decimal(10, 2)	
category_mandates	uuid[]	ARR
geography_focus	varchar[]	ARR
min_ticket_usd_mn	decimal(10, 2)	
esg_priority	bool	
contact_json	jsonb	
relations: Portfolio → Deal → AlertRule		

Brand & Product

The investable asset layer: brand financial KPIs, product line sell-through, price architecture, channel mix, and competitive positioning.

Brand

Fashion label or operator entity — the core investable unit. Stores financials, brand equity score, ownership type, and sustainability certifications.

field	type	tag
brand_id	uuid	PK
name	varchar(120)	
category_id	uuid	FK
founded_year	int	
hq_country_iso	char(3)	
annual_revenue_usd	decimal(16, 2)	
revenue_growth_pct	decimal(6, 2)	CALC
gross_margin_pct	decimal(5, 2)	
ebitda_margin_pct	decimal(5, 2)	
brand_equity_score	decimal(5, 2)	CALC
ownership_type	enum	
certification_ids	uuid[]	ARR
relations: ProductLine → PricingTier → DistributionChannel → CompetitorProfile		

ProductLine			
SKU group with category, season, and performance tracking. Captures sell-through rate, return rate, and sustainability tier per line.			
field	type		tag
line_id	uuid		PK
brand_id	uuid		FK
name	varchar(100)		
category_id	uuid		FK
season_code	varchar(10)		IDX
sku_count	int		
avg_asp_usd	decimal(10, 2)		
sell_through_rate	decimal(4,3)		CALC
return_rate	decimal(4,3)		
sustainability_tier	enum		
materials_json	jsonb		
relations: PricingTier → SentimentSignal → TrendIndicator			

PricingTier			
Price architecture segment (mass, accessible luxury, ultra-luxury) with ASP range, elasticity coefficient, and markdown depth.			
field	type		tag
tier_id	uuid		PK
brand_id	uuid		FK
tier_label	enum		
asp_floor_usd	decimal(10, 2)		
asp_ceiling_usd	decimal(10, 2)		
elasticity_coeff	decimal(6,4)		CALC
markdown_depth_pct	decimal(5,2)		
effective_from	date		
relations: MarketSegment → ConsumerSegment			

DistributionChannel

Sales channel mix per brand — DTC, wholesale, marketplace, retail — with CAC, LTV, and GMV tracking.

field	type	tag
channel_id	uuid	PK
brand_id	uuid	FK
channel_type	enum	
platform_name	varchar(80)	
geography_iso	char(3)	
revenue_share_pct	decimal(5,2)	
cac_usd	decimal(8,2)	CALC
ltv_usd	decimal(10,2)	CALC
active_skus	int	
gmv_usd	decimal(14,2)	
relations: SocialListening → ConsumerSegment		

CompetitorProfile

Competitive set mapping with share-of-voice, pricing gap, and growth differential against comparable brands.

field	type	tag
competitor_id	uuid	PK
brand_id	uuid	FK
competitor_brand_id	uuid	FK
overlap_score	decimal(4,3)	CALC
share_of_voice_pct	decimal(5,2)	CALC
pricing_gap_pct	decimal(6,2)	CALC
growth_delta_pct	decimal(6,2)	CALC
as_of_date	date	IDX
relations: Benchmark → InvestmentScore		

Consumer & Sentiment

Demand-side intelligence: psychographic segments, NLP-derived sentiment aggregation, and platform-level social listening with viral coefficient tracking.

ConsumerSegment

Psychographic or demographic cohort with spending profile, fashion affinity score, sustainability index, and channel preference mix.

field	type	tag
segment_id	uuid	PK
name	varchar(100)	
age_band	int4range	
income_band_usd	numrange	
geography_iso	char(3)	IDX
fashion_affinity_score	decimal(4,2)	CALC
digital_native	bool	
sustainability_index	decimal(4,2)	
avg_annual_spend_usd	decimal(10,2)	
channel_preferences	jsonb	
relations: SentimentSignal → SocialListening → TrendIndicator		

SentimentSignal

Aggregated NLP-derived sentiment scores from reviews, forums, and press. Compound score with positive/negative ratio and mention volume.

field	type	tag
signal_id	uuid	PK
brand_id	uuid	FK
segment_id	uuid	FK
source_type	enum	
compound_score	decimal(5,4)	CALC
positive_ratio	decimal(4,3)	CALC
negative_ratio	decimal(4,3)	CALC
volume_mentions	int	
period_start	date	IDX
period_end	date	
model_id	varchar(40)	
relations: MarketSignal → TrendIndicator → AlertRule		

SocialListening		
Platform-level volume, velocity, and viral moment tracking. Captures impressions, engagement, share-of-voice, and influencer tier mix.		
field	type	tag
listening_id	uuid	PK
brand_id	uuid	FK
platform	enum	IDX
hashtag_cluster	varchar[]	ARR
impressions	bigint	
engagements	bigint	
share_of_voice_pct	decimal(5,2)	CALC
influencer_tier_mix	jsonb	
viral_coefficient	decimal(6,4)	CALC
captured_at	timestampz	IDX
relations: SentimentSignal → TrendIndicator → MarketSignal		

Intelligence Engine

The scoring, forecasting, alerting, and reporting layer. Aggregates all upstream signals into versioned InvestmentScores, ML-driven forecasts, and audit-ready snapshots.

InvestmentScore

Composite investability rating per brand or category. Weighted sub-scores: momentum, brand equity, financial health, ESG, and risk. Versioned by model run.

field	type	tag
score_id	uuid	PK
brand_id	uuid	FK
category_id	uuid	FK
overall_score	decimal(5,2)	CALC
momentum_score	decimal(5,2)	CALC
brand_equity_score	decimal(5,2)	CALC
financial_health_score	decimal(5,2)	CALC
esg_score	decimal(5,2)	CALC
risk_score	decimal(5,2)	CALC
model_version	varchar(20)	
weight_config	jsonb	
computed_at	timestampz	IDX
relations: AlertRule → ReportSnapshot → Benchmark		

AlertRule

Configurable threshold rules firing notifications or workflow actions. Supports cooldown periods, multi-channel delivery, and per-portfolio scoping.

field	type	tag
rule_id	uuid	PK
portfolio_id	uuid	FK
name	varchar(100)	
entity_type	enum	
condition_json	jsonb	
severity	enum	
channels	varchar[]	ARR
cooldown_minutes	int	
is_active	bool	
last_fired_at	timestampz	
relations: ReportSnapshot → InvestorProfile		

ForecastModel

Time-series and ML model run for revenue, TAM, or score projection. Stores predictions, confidence intervals, feature importance, and accuracy metrics.

field	type	tag
model_run_id	uuid	PK
model_type	enum	
target_entity_type	enum	
target_entity_id	uuid	
horizon_months	int	
predictions	jsonb	
confidence_intervals	jsonb	
feature_importance	jsonb	
rmse	decimal(10, 4)	CALC
mape_pct	decimal(6,3)	CALC
ran_at	timestamp z	IDX
relations: ReportSnapshot → InvestmentScore → Benchmark		

ReportSnapshot

Versioned investment memo or dashboard snapshot for full audit trail. Stores structured JSON content and a linked PDF artefact in object storage.

field	type	tag
snapshot_id	uuid	PK
report_type	enum	
portfolio_id	uuid	FK
deal_id	uuid	FK
generated_by	uuid	FK
content_json	jsonb	
pdf_s3_key	varchar(255)	
status	enum	
generated_at	timestamp z	IDX
approved_at	timestamp z	
relations: InvestorProfile		

Benchmark		
Industry or peer-group reference metrics (P25/P50/P75/P90) for relative valuation. Supports EV/Revenue, EV/EBITDA, and growth multiples.		
field	type	tag
benchmark_id	uuid	PK
category_id	uuid	FK
metric_name	varchar(80)	
peer_group_ids	uuid[]	ARR
p25_value	decimal(12, 4)	
p50_value	decimal(12, 4)	
p75_value	decimal(12, 4)	
p90_value	decimal(12, 4)	
as_of_date	date	IDX
source	varchar(80)	
relations: InvestmentScore → Valuation → ForecastModel		